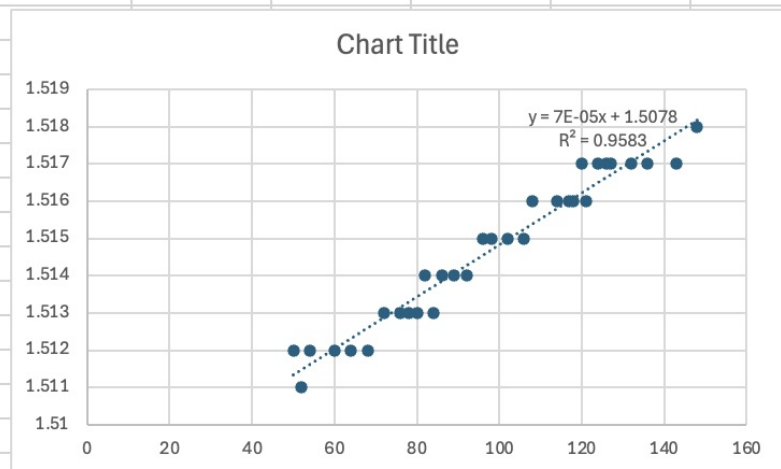


Free oscillation

Amplitude θ	Period T_0 (s)
148	1.518
143	1.517
136	1.517
132	1.517
127	1.517
126	1.517
124	1.517
121	1.516
120	1.517
118	1.516
117	1.516
114	1.516
108	1.516
106	1.515
102	1.515
98	1.515
96	1.515
92	1.514
89	1.514
86	1.514
84	1.513
82	1.514
80	1.513
78	1.513
76	1.513
72	1.513
68	1.512
64	1.512
60	1.512
54	1.512
52	1.511
50	1.512



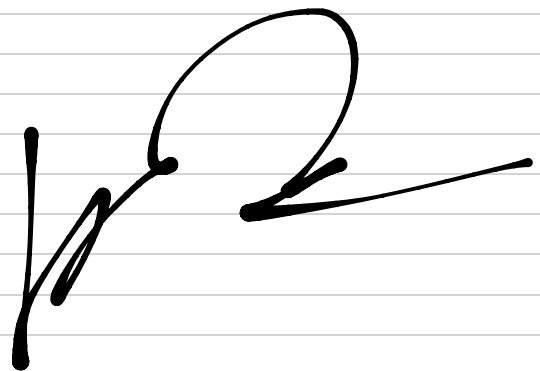
Damped oscillations

Amplitude θ	Period T_0 (s)	Amplitude θ	Period T_0 (s)	Amplitude θ	Period T_0 (s)
132	15.142	140	15.141	118	15.136
120	15.142	104	15.141	106	15.136
110	15.142	94	15.141	95	15.136
100	15.142	84	15.141	86	15.136
91	15.142	76	15.141	76	15.136
82	15.142	68	15.141	68	15.136
75	15.142	62	15.141	62	15.136
68	15.142	56	15.141	55	15.136
62	15.142	50	15.141	49	15.136
56	15.142	45	15.141	44	15.136
Tbar	15.142		15.141		15.136
beta	0.00566273		0.00749607		0.00651754
	Damp1		Damp2		Damp3

Forced oscillations

Motor period T (s)	Phase shift ϕ_2 (°)	Amplitude θ (°)	Motor period T (s)	Phase shift ϕ_2 (°)	Amplitude θ (°)
15.139	100.0	154	15.140	99.5	142
15.177	90.0	158	15.176	91.0	145
15.232	79.5	156	15.217	80.0	144
15.261	68.5	150	15.292	70.0	133
15.216	60.5	156	15.315	60.0	128
15.347	52.0	126	15.361	50.0	114
15.407	42.0	108	15.444	40.0	95
15.499	32.0	86	15.546	30.0	76
15.660	21.0	63	15.733	20.0	56
15.969	10.0	42	16.225	10.0	34
Damp 1			Damp 2		

Signature:



PHYS 121 – Lab Report: Pohl’s Pendulum

Juntang Wang

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Date:	November 12, 2025
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Instructor:	Kai Wang and Chen Xi
Section:	Wednesday

1 Revisit the Theory Behind Different Types of Oscillations

1.1 Free Oscillation

Based on Newton’s second law for rotation, the differential equation describing free oscillation is:

$$I \frac{d^2\theta}{dt^2} = -k\theta \quad (1)$$

where I is the moment of inertia of the pendulum and k is the stiffness coefficient of the spiral spring.

One solution to this differential equation is a sinusoidal function:

$$\theta(t) = \theta_0 \cos(\omega_0 t - \phi) \quad (2)$$

where θ_0 is the oscillation amplitude, $\omega_0 = \sqrt{k/I}$ is the angular frequency of the free oscillator, and ϕ is the phase shift/initial phase.

Assumptions needed to obtain these results:

- Small angle approximation: The restoring torque is proportional to the displacement angle θ (Equation 1: $M_s = -k\theta$).
- No damping: There are no frictional or drag forces acting on the pendulum ($b = 0$).
- No external forces: The system is not subject to any periodic external driving torque.
- The spiral spring provides a linear restoring torque proportional to the angular displacement.

1.2 Damped Oscillation

When frictional/drag forces act on the pendulum, the differential equation becomes:

$$I \frac{d^2\theta}{dt^2} = -k\theta - b \frac{d\theta}{dt} \quad (3)$$

or equivalently:

$$\frac{d^2\theta}{dt^2} + \frac{b}{I} \frac{d\theta}{dt} + \frac{k}{I} \theta = 0 \quad (4)$$

where b is the friction/drag parameter.

For weak damping, one solution to this differential equation is:

$$\theta(t) = \theta_1 e^{-\beta t} \cos(\omega_1 t - \phi_1) \quad (5)$$

where $\beta = b/2I$ is the damping coefficient, θ_1 is the maximum amplitude, $\omega_1 = \sqrt{\omega_0^2 - \beta^2}$ is the angular frequency of the damped oscillation, and ϕ_1 is the phase constant.

Assumptions needed to obtain these results:

- Weak damping: The damping coefficient satisfies $\beta < \omega_0$, ensuring the motion remains oscillatory (not aperiodic).
- Linear damping: The damping torque is proportional to the angular velocity ($M_d = -b d\theta/dt$).
- Small angle approximation: The restoring torque remains proportional to displacement.
- The damping is electromagnetic in nature (from the damping coil), producing a torque proportional to angular velocity.

1.3 Forced Oscillation

When an additional periodic external torque acts on the pendulum, the differential equation becomes:

$$I \frac{d^2\theta}{dt^2} = -k\theta - b \frac{d\theta}{dt} + M_0 \cos(\omega_2 t) \quad (6)$$

or equivalently:

$$\frac{d^2\theta}{dt^2} + 2\beta \frac{d\theta}{dt} + \omega_0^2 \theta = F_0 \cos(\omega_2 t) \quad (7)$$

where $F_0 = M_0/I$ is the magnitude of the external force and ω_2 is the angular frequency of the driving torque.

The complete solution consists of two parts:

$$\theta(t) = \theta_1 e^{-\beta t} \cos(\omega_1 t - \phi_1) + \theta_2 \cos(\omega_2 t - \phi_2) \quad (8)$$

After a long time ($t \gg \beta^{-1}$), the damped oscillation decays, and the system oscillates at the driving frequency:

$$\theta(t) = \theta_2 \cos(\omega_2 t - \phi_2) \quad (9)$$

where the amplitude θ_2 and phase shift ϕ_2 are given by:

$$\theta_2 = \frac{F_0}{\sqrt{(\omega_0^2 - \omega_2^2)^2 + 4\beta^2\omega_2^2}} \quad (10)$$

$$\phi_2 = \tan^{-1} \left(\frac{2\beta\omega_2}{\omega_0^2 - \omega_2^2} \right) = \tan^{-1} \left(\frac{\beta T_0^2 T}{\pi(T^2 - T_0^2)} \right) \quad (11)$$

Assumptions needed to obtain these results:

- Steady state reached: The measurement is taken after sufficient time has elapsed ($t \gg \beta^{-1}$) so that the transient damped oscillation has decayed.
- Weak damping: The system remains oscillatory ($\beta < \omega_0$).
- Periodic driving force: The external torque is sinusoidal with constant amplitude M_0 and frequency ω_2 .
- Small angle approximation: The restoring torque remains linear.
- The motor produces a periodic torque whose frequency can be precisely controlled.

2 Data Analysis and Discussion

2.1 Free Oscillation

Table 1 shows the correlation between the natural period T_0 and the amplitude θ_0 of free oscillation.

Figure 1 shows the relationship between the natural period T_0 and the amplitude θ_0 . A linear fit was performed to determine the correlation between these quantities.

A linear fit of the form $T_0 = a\theta_0 + b$ yields $T_0 = 1.5080 + 6.95 \times 10^{-5}\theta_0$ (with θ_0 in degrees). The linear fit shows a weak positive correlation between amplitude and period, with the period increasing slightly as amplitude increases. This suggests that the system exhibits some amplitude dependence, likely due to nonlinear effects that become more significant at larger amplitudes, violating the small angle approximation.

2.2 Damped Oscillation

The damping coefficient β was calculated using Equation (20):

$$5\beta\bar{T} = \ln \left(\frac{\theta_i}{\theta_{i+5}} \right) \quad (12)$$

where $\bar{T}$ is the average period and $i = 1, 2, \dots, 5$.

Tables 2, 3, and 4 show the amplitude measurements and calculations for the three damping levels.

The damping coefficients calculated are: $\beta_1 = 0.00566273 \text{ s}^{-1}$ (Damp1), $\beta_2 = 0.00749607 \text{ s}^{-1}$ (Damp2), and $\beta_3 = 0.00651754 \text{ s}^{-1}$ (Damp3). As expected, Damp2 shows the highest damping coefficient, indicating stronger electromagnetic damping. The amplitude decay follows an exponential pattern. The decay rate increases with the damping coefficient.

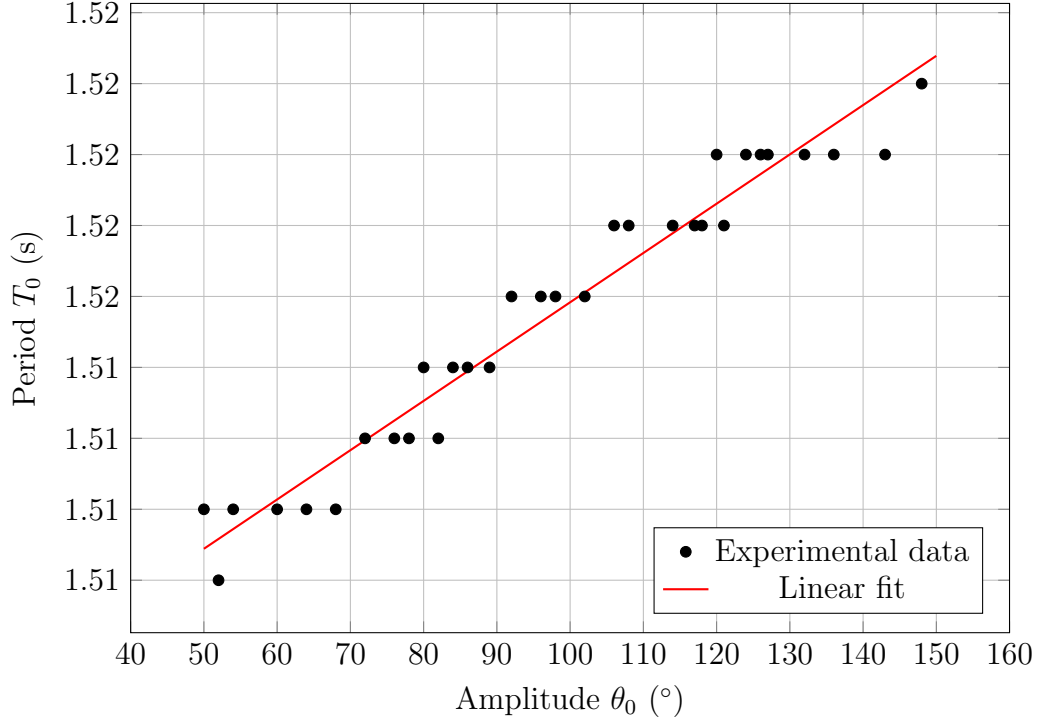


Figure 1: Natural period T_0 as a function of amplitude θ_0 for free oscillation

2.3 Forced Oscillation

Tables 5 and 6 show the measurements of amplitude and phase shift for forced oscillations at different motor periods. The corresponding T_0 values were determined from the linear fit of the free oscillation data, and ω/ω_r and ϕ'_2 were calculated using Equations (17) and (16).

Note: The T_0 values in the tables are determined from the linear fit of free oscillation data: $T_0 = 1.5080 + 6.95 \times 10^{-5} \theta$ (with θ in degrees). The ω/ω_r values are calculated using $\omega/\omega_r = T_0/T$, where T is the motor period and T_0 is the natural period corresponding to the measured amplitude. The ϕ'_2 values are calculated using Equation (16): $\phi'_2 = \tan^{-1}(\beta T_0^2 T / [\pi(T^2 - T_0^2)])$, where β is the damping coefficient for the respective damping level.

Figure 2 shows the resonance curves for both damping levels, plotting amplitude θ as a function of ω/ω_r (displayed in units of 10^{-2}).

The resonance curves show that the amplitude reaches a maximum near $\omega/\omega_r \approx 1$ (resonance condition). Damp1 exhibits a higher peak amplitude than Damp2, consistent with the lower damping coefficient for Damp1. As the driving frequency deviates from the natural frequency, the amplitude decreases.

Figure 3 shows the comparison between the measured phase shift ϕ_2 and the calculated phase shift ϕ'_2 as a function of ω/ω_r (displayed in units of 10^{-2}).

The phase shift plot shows that the measured phase shifts ϕ_2 for both damping levels intersect near resonance ($\omega/\omega_r \approx 1$), as expected from theory. At resonance, both damping levels exhibit similar phase shifts, with the intersection point clearly visible in the plot. The calculated phase shifts ϕ'_2 remain relatively constant near 89° for both damping levels, consistent with the theoretical prediction. The measured phase shifts decrease as ω/ω_r

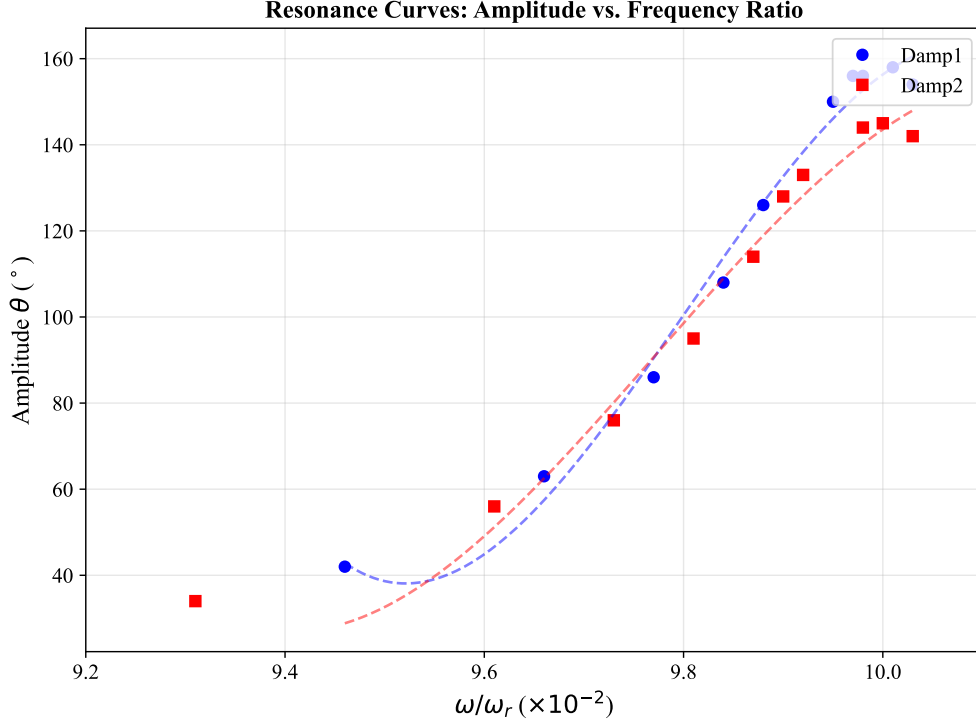


Figure 2: Resonance curves: Amplitude θ as a function of $\omega/\omega_r (\times 10^{-2})$ for different damping levels

decreases from resonance, showing good agreement with the expected behavior.

2.4 Error Analysis and Discussion

Several sources of experimental error and limitations affect the results:

Sources of Error:

1. **Measurement uncertainty:** The photogate measurements have finite resolution, and reading the phase shift using the stroboscopic effect introduces visual estimation errors.
2. **Systematic errors:** The assumption that T_0 is constant for all amplitudes may not be valid, as the free oscillation data shows a slight dependence of period on amplitude. This nonlinearity violates the small angle approximation.
3. **Steady state assumption:** For forced oscillations, the system may not have fully reached steady state when measurements are taken, especially when the motor period is changed frequently.
4. **Damping level consistency:** The damping coefficient may not be perfectly constant throughout the experiment, as the electromagnetic damping depends on the current through the coil, which may fluctuate.

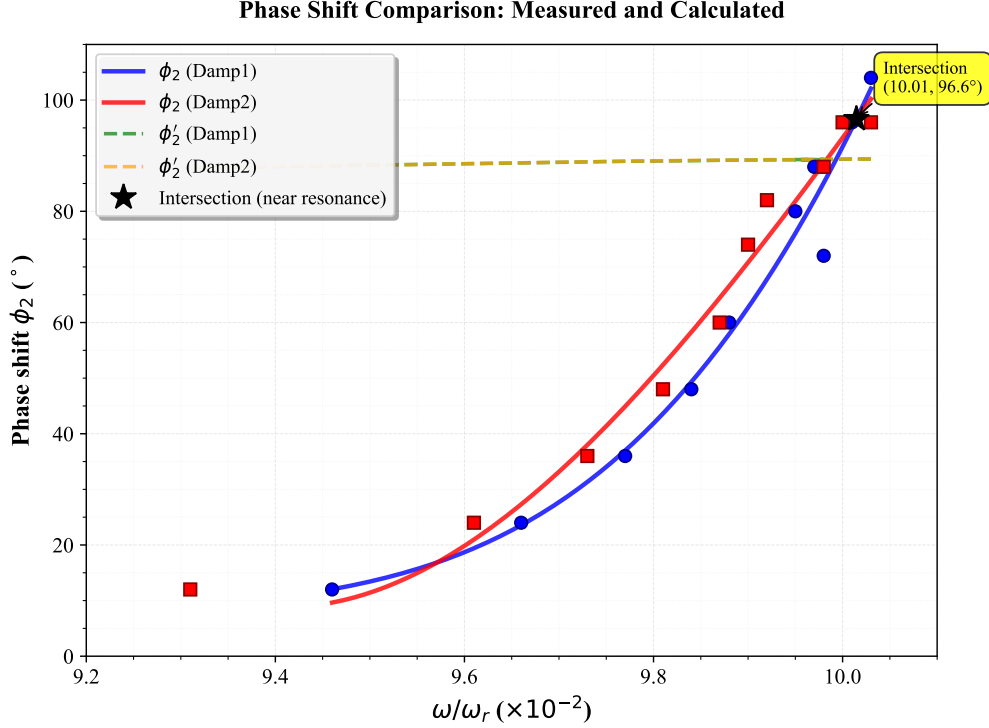


Figure 3: Phase shift comparison: Measured ϕ_2 and calculated ϕ'_2 as a function of ω/ω_r ($\times 10^{-2}$). The intersection point between the measured phase shift curves for Damp1 and Damp2 is indicated, occurring near resonance.

5. **Phase shift measurement:** The stroboscopic effect method for measuring phase shift relies on visual observation, which introduces subjective errors. The flash lamp may also interfere with photogate measurements.
6. **Amplitude-dependent period:** The free oscillation data shows that the period increases slightly with amplitude, indicating nonlinear effects that are not accounted for in the linear theory.

Comparison with Theoretical Predictions:

The experimental results show some discrepancies with theoretical predictions:

- The free oscillation period shows a weak dependence on amplitude, which contradicts the assumption of a constant natural frequency. This suggests that the small angle approximation breaks down at larger amplitudes.
- The damping coefficients calculated from the amplitude decay are consistent with the expected behavior, showing exponential decay as predicted by theory.
- The resonance curves show the expected behavior with maximum amplitude near $\omega/\omega_r \approx 1$, but the peak is broader than predicted, possibly due to the amplitude-dependent natural frequency.

- The phase shift measurements show significant deviation from the calculated values, particularly at frequencies away from resonance. This may be due to the limitations of the stroboscopic measurement method or the breakdown of the linear theory assumptions.

Discussion of Assumptions:

The assumptions made in the theory sections are not always valid in practice:

- **Small angle approximation:** At larger amplitudes (above 100°), the restoring torque may not be perfectly linear, leading to amplitude-dependent period.
- **Linear damping:** The electromagnetic damping is approximately linear for small velocities, but may show nonlinear behavior at higher oscillation speeds.
- **Steady state:** The requirement that $t \gg \beta^{-1}$ may not always be satisfied, especially when the motor period is adjusted frequently.
- **Constant natural frequency:** The free oscillation data shows that the natural frequency depends on amplitude, violating the assumption of a constant ω_0 .

2.5 Questions

Question 1: Can we use the data obtained for free oscillation to calculate the damping coefficient β with Equation (20)? If yes, please show it. If no, why not?

Answer: No, we cannot use the free oscillation data to calculate the damping coefficient β using Equation (20).

Free oscillation, by definition, occurs when there is no damping ($b = 0$), which means $\beta = b/2I = 0$. Equation (20) is derived from the damped oscillation solution, which assumes $\beta > 0$. In free oscillation, the amplitude remains constant (in the ideal case) or decreases very slowly due to minimal friction, but this decay is not described by the exponential damping term $e^{-\beta t}$ that appears in Equation (8) for damped oscillations.

To calculate β , we need data from damped oscillations where the amplitude decreases exponentially over time, allowing us to measure the decay rate. The free oscillation experiment is specifically designed to minimize damping, so it cannot provide meaningful data for calculating the damping coefficient.

Question 2: Give two simple examples of the resonance phenomenon, one of them is good for our life and give a solution on how to make use of it, while the other example is bad for our life and give a solution on how to avoid it.

Answer:

Beneficial Resonance Example: Musical Instruments

Musical instruments, such as guitars and violins, rely on resonance to amplify sound. When a string is plucked, it vibrates at its natural frequency. The body of the instrument is designed to resonate at frequencies that match the harmonics of the strings, amplifying the sound and creating rich, full tones.

How to make use of it: Instrument makers carefully design the shape, size, and material of the instrument body to have specific resonant frequencies that enhance the desired musical frequencies. Musicians can also use resonance by placing instruments in acoustically

favorable environments (concert halls) that enhance certain frequencies through constructive interference.

Harmful Resonance Example: Mechanical Vibrations in Structures

When external forces (such as wind, earthquakes, or traffic) act on structures at frequencies matching their natural frequencies, resonance can cause excessive vibrations. This was dramatically demonstrated in the 1940 collapse of the Tacoma Narrows Bridge, where wind-induced oscillations at the bridge's natural frequency caused catastrophic failure.

How to avoid it: Engineers use several strategies:

- **Damping systems:** Install tuned mass dampers or viscous dampers that absorb vibrational energy, similar to the damping coil in Pohl's pendulum.
- **Structural design:** Modify the natural frequency of the structure by changing its mass or stiffness so it doesn't match common driving frequencies (e.g., typical wind frequencies or earthquake frequencies).
- **Active control systems:** Use sensors and actuators to detect and counteract resonant vibrations in real-time.
- **Material selection:** Use materials with high internal damping (energy dissipation) to reduce resonance effects.

3 Conclusion

This experiment successfully investigated the three types of oscillations in Pohl's pendulum: free, damped, and forced. The free oscillation data revealed a slight amplitude dependence of the period, indicating nonlinear effects at larger amplitudes. The damped oscillation measurements allowed calculation of damping coefficients for three different damping levels, showing exponential amplitude decay as predicted by theory. The forced oscillation experiments demonstrated resonance behavior, with maximum amplitude occurring when the driving frequency matched the natural frequency. However, discrepancies between measured and calculated phase shifts suggest limitations in the linear theory assumptions, particularly the small angle approximation and constant natural frequency assumption.

The experiment highlights the importance of understanding the assumptions underlying theoretical models and recognizing when these assumptions break down in real systems. The resonance phenomenon observed has both beneficial applications (musical instruments) and potentially harmful consequences (structural vibrations), demonstrating the practical importance of understanding oscillatory behavior.

Additional Notes

Large language models (LLMs) were used as translators and brainstorming assistants during the completion of this report, no other additional help was provided.

Table 1: Correlation between the natural period T_0 and the amplitude θ_0 of free oscillation

Amplitude θ_0 ($^\circ$)	Period T_0 (s)
148	1.518
143	1.517
136	1.517
132	1.517
127	1.517
126	1.517
124	1.517
121	1.516
120	1.517
118	1.516
117	1.516
114	1.516
108	1.516
106	1.516
102	1.515
98	1.515
96	1.515
92	1.515
89	1.514
86	1.514
84	1.514
82	1.513
80	1.514
78	1.513
76	1.513
72	1.513
68	1.512
64	1.512
60	1.512
54	1.512
52	1.511
50	1.512

Table 2: Amplitude of different oscillation periods in the damped mode (Damp1)

Amplitude #	Amplitude θ ($^\circ$)	Amplitude #	Amplitude θ ($^\circ$)	$\ln(\theta_i/\theta_{i+5})$
θ_1	132	θ_6	90	0.3838
θ_2	120	θ_7	78	0.4308
θ_3	108	θ_8	68	0.4620
θ_4	98	θ_9	60	0.4904
θ_5	88	θ_{10}	56	0.4511
Average of $\ln(\theta_i/\theta_{i+5})$				0.4436
Damping coefficient β (Equation 20)				0.005 662 73
$10\bar{T} = 151.42$ s; $\bar{T} = 15.142$ s				

Table 3: Amplitude of different oscillation periods in the damped mode (Damp2)

Amplitude #	Amplitude θ ($^\circ$)	Amplitude #	Amplitude θ ($^\circ$)	$\ln(\theta_i/\theta_{i+5})$
θ_1	140	θ_6	88	0.4659
θ_2	126	θ_7	76	0.5073
θ_3	114	θ_8	66	0.5481
θ_4	104	θ_9	58	0.5827
θ_5	94	θ_{10}	45	0.7309
Average of $\ln(\theta_i/\theta_{i+5})$				0.5670
Damping coefficient β (Equation 20)				0.007 496 07
$10\bar{T} = 151.41$ s; $\bar{T} = 15.141$ s				

Table 4: Amplitude of different oscillation periods in the damped mode (Damp3)

Amplitude #	Amplitude θ ($^\circ$)	Amplitude #	Amplitude θ ($^\circ$)	$\ln(\theta_i/\theta_{i+5})$
θ_1	118	θ_6	80	0.3894
θ_2	108	θ_7	72	0.4055
θ_3	98	θ_8	64	0.4253
θ_4	90	θ_9	58	0.4377
θ_5	82	θ_{10}	44	0.6233
Average of $\ln(\theta_i/\theta_{i+5})$				0.4562
Damping coefficient β (Equation 20)				0.006 517 54
$10\bar{T} = 151.36$ s; $\bar{T} = 15.136$ s				

Table 5: Amplitude θ and phase shift of forced oscillation with different natural period T_0 of the pendulum (Damp1)

Measurement			Calculation (After lab)		
Motor period T (s)	Phase shift ϕ_2 ($^\circ$)	Amplitude θ ($^\circ$)	T_0 (s)	ω/ω_r	ϕ_2' ($^\circ$)
15.139	100.0	154	1.519	0.1003	89.4
15.177	90.0	158	1.519	0.1001	89.4
15.232	79.5	156	1.519	0.0997	89.3
15.261	68.5	150	1.518	0.0995	89.3
15.216	60.5	156	1.519	0.0998	89.3
15.347	52.0	126	1.517	0.0988	89.2
15.407	42.0	108	1.515	0.0984	89.1
15.499	32.0	86	1.514	0.0977	89.0
15.660	21.0	63	1.512	0.0966	88.7
15.969	10.0	42	1.511	0.0946	88.2

Table 6: Amplitude θ and phase shift of forced oscillation with different natural period T_0 of the pendulum (Damp2)

Measurement			Calculation (After lab)		
Motor period T (s)	Phase shift ϕ_2 ($^\circ$)	Amplitude θ ($^\circ$)	T_0 (s)	ω/ω_r	ϕ_2' ($^\circ$)
15.140	99.5	142	1.518	0.1003	89.4
15.176	91.0	145	1.518	0.1000	89.4
15.217	80.0	144	1.518	0.0998	89.3
15.292	70.0	133	1.517	0.0992	89.3
15.315	60.0	128	1.517	0.0990	89.2
15.361	50.0	114	1.516	0.0987	89.2
15.444	40.0	95	1.515	0.0981	89.1
15.546	30.0	76	1.513	0.0973	88.9
15.733	20.0	56	1.512	0.0961	88.6
16.225	10.0	34	1.510	0.0931	87.7